

On Power Domination Problem of Product Graphs *

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Abstract

Phase measurement units (PMU) placement problem is an important task for electric power company. This problem can be modeled by graphs as follows: Given an undirected graph $G = (V, E)$, find a set $S \subseteq V(G)$ with minimum cardinality such that all vertices in V are observed by some vertices in S . Herein, a vertex which is claimed to be a PMU observes itself and all its neighbors, and if an observed vertex has all but one of its neighbors observed, then the remaining neighbor becomes observed as well. The minimum cardinality of S for graph G is called power domination number and is denoted as $\gamma_p(G)$. This paper established the power domination number of tensor product, and strong product of some graphs.

1 Introduction

The power domination problem arose in the context of observing electric power system. A power system contains a set of nodes and a set of links connecting the nodes. It also contains a set of generators which supply power, and a set of loads where the power is directed to[1]. It is important for electric power companies to continuously monitoring the power systems and observing all the states, such as the voltage magnitude of loads and the current phase measurements at branches. A *Phase Measurement Unit* (PMU) is a measurement device placed on a node that has the ability to measure the voltage of the node and current phase of the links connected to the nodes, and to give warnings of system-wide failures.

In order to observe all the state variables of the system, they need to place PMUs at selected locations in the system. A system is said to be *observed* if all of the state variables of the system can be determined from a set of measurements (e.g., voltages, currents, and phase angle).

One can increase the observed area of the system by increasing the number of PMUs. Because of

the high cost of a PMU, it is very important to determine an optimal allocation in order to minimize the number of PMUs required while maintaining the ability to observe (monitor) the entire system.

An electric power system can be interpreted as an undirected graph in a very natural fashion. Let $G = (V, E)$ be a graph representing an electric power system where a vertex represents an electrical node (a measuring point) and an edge represents a transmission line joining two electrical nodes. We may observe the voltage and phase angle at some measuring point, directly by placing a PMU on the vertex, or observe measuring points indirectly by using the propagation rules which are derived from the Ohms and Kirchoffs laws for an electric circuit[1].

The problem of locating a smallest set of PMUs to observe the entire system is a graph theoretical problem closely related to the well-known vertex cover and domination problems. Hence, this problem is not only interested in the power system industry but also interested in the research of graph theory. However, this type of domination is different from the standard domination problem since the domination rules can be iterated here.

2 Terminologies and Related Works

For a graph $G = (V, E)$, let $v \in V(G)$. The *open neighborhood* of v , denoted as $N_G(v)$, is the set of vertices adjacent with v ; and the *closed neighborhood* of v , denoted as $N_G[v]$, is $N_G(v) \cup \{v\}$. For a set $S \subseteq V(G)$, the *open neighborhood* of S is defined as $N_G(S) = \bigcup_{v \in S} N_G(v) - S$ and the *closed neighborhood* of S is defined as $N_G[S] = N_G(S) \cup S$. When it is not confusion, we may omit G in the notation of neighborhood.

A set $S \subseteq V$ is a *dominating set* of G if every vertex not in S has a neighbor in S . The power domination may be considered as a variation of traditional domination. A set $S \subseteq V$ is a *power dominating set* (PDS) of G if every vertex not in S is power dominated by S using the power domination rules as follows:

1. A set S can power dominate $N[S]$.

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2. If a power dominated vertex has $k > 1$ adjacent vertices and $k - 1$ of these vertices are power dominated, then all k vertices are power dominated.

The minimum cardinality among all PDS of G is the *power domination number* and is denoted as $\gamma_p(G)$. From now on, when there is no confusion, this thesis uses the term *domination* to substitute *power domination*.

The decision version of the power domination problem has been shown to be NP-complete even when the graphs are restricted to bipartite graphs and chordal graphs [8], to planar cubic graphs [18], to circle graphs [7][23], and to split graphs [13][23]. A variety of heuristic approaches to find power domination number for general graphs or some special graphs have been developed in the past two decades [7][10][11]. The power domination problem has efficient polynomial time algorithms for the classes of graphs such as trees [8], graphs with bounded treewidth [7], block graphs [9][22], block-cactus graphs [9], interval [13], grids [16], honeycomb meshes [20] and circular-arc graphs [14].

Most of the formula solutions of power domination number were given on product graphs. Dorbec et al. [5] determined the power domination number for all tensor product of paths except of the odd component of the tensor product of two odd paths (which they provided an upper bound). They also computed the same for the strong product of paths and the lexicographic product of any two graphs. Varghese et al. [19] provided a formula solution for the tensor product of an even path with a cycle and for Mycielskian of the complete graph, wheel, n -fan and n -star.

For special graphs, the upper bounds of $\gamma_p(G)$ are given for cylinder, torus[2], for hypercube [4][15], and for generalized Petersen graphs [2][12][21]. The upper bounds on the power domination number are also given for block graphs [22], for a connected graph with at least three vertices, and for a connected claw-free cubic graph [24]. Moreover, [3] and [24] independently showed that the power domination number in a connected graph with $n \geq 3$ vertices is no more than $n/3$, and characterized the extremal graphs that attain the upper bound.

Pai et al. [17] investigated the restricted and fault tolerant power domination problems and determined the power domination numbers in grid graphs. This paper established the power domination number of tensor product of a cycle with a cycle, and strong product of a cycle with either a cycle or a path.

3 Tensor Product

The tensor product graph $C_m \times C_n$ of two cycles C_m and C_n for $m, n \geq 3$, is a graph with vertex set $V(C_m \times C_n) = \{v_{i,j} | 0 \leq i < m, 0 \leq j < n\}$ and edge set $E(C_m \times C_n) = \{v_{i,j}v_{i+1,j+1}, v_{i,j}v_{i+1,j-1}, v_{i,j}v_{i-1,j+1}, v_{i,j}v_{i-1,j-1}\}$ where $0 \leq i \leq m-1, 0 \leq j \leq n-1$ and the subscript sum is taken modulo m or n . For convenience, let V_{odd} and V_{even} denoted the vertex sets such that $V_{\text{odd}} = \{v_{i,j} | (i+j) \equiv 1 \pmod{2}\}$ and $V_{\text{even}} = \{v_{i,j} | (i+j) \equiv 0 \pmod{2}\}$ respectively. An example of $C_3 \times C_4$ is illustrated in Figure 1.

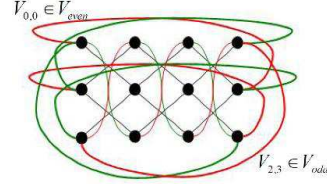


Figure 1: An illustration of $C_3 \times C_4$.

One way notice that when $G = C_m \times C_n$ for some odd n . A vertex $v_{i,j} \in V_{\text{odd}}$ with $i < m$ and $j < n$ can be reconsidered as $v_{i,j+n}$, which is a vertex in V_{even} . When $G = C_m \times C_n$ for some odd m . A vertex $v_{i,j} \in V_{\text{odd}}$ with $i < m$ and $j < n$ can be reconsidered as $v_{i+m,j}$, which is a vertex in V_{even} .

Lemma 1 shows the dominating area of a set S may be increased several columns and rows by adding just one more vertex to S .

Lemma 1 Let $G = C_m \times C_n$.

- (1) If $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq a, 0 \leq j \leq b\}$ where $(a+b) \equiv 0 \pmod{2}$ is power dominated by S , then $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq a+1, 0 \leq j \leq b+3\}$ is power dominated by $S' = S \cup \{v_{a,b+2}\}$; and $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq a+3, 0 \leq j \leq b+1\}$ is power dominated by $S'' = S \cup \{v_{a+2,b}\}$.
- (2) If $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq a, 0 \leq j \leq b\}$ where $(a+b) \equiv 1 \pmod{2}$ is power dominated by S , then $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq a+1, 0 \leq j \leq b+3\}$ is power dominated by $S' = S \cup \{v_{a,b+2}\}$; and $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq a+3, 0 \leq j \leq b+1\}$ is power dominated by $S'' = S \cup \{v_{a+2,b}\}$.

Proof. Because of the symmetrization, without loss of generality this proof only consider S' of (1). Let $V_k = \{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq a, 0 \leq j \leq b\}$ be power dominated by $S_k \subseteq V(G)$. Consider $S_{k+1} = S_k \cup \{v_{a,b+2}\}$. Since $N(v_{a,b+2}) = \{v_{a-1,b+1}, v_{a+1,b+1}, v_{a-1,b+3}, v_{a+1,b+3}\}$, by rule 1, the set $V_k \cup \{v_{a-1,b+1}, v_{a+1,b+1}, v_{a-1,b+3}, v_{a+1,b+3}\}$

is power dominated by S_{k+1} . Since $N(v_{a,b}) = \{v_{a-1,b-1}, v_{a-1,b+1}, v_{a+1,b-1}, v_{a+1,b+1}\}$ and $v_{a-1,b-1}, v_{a-1,b+1}$, and $v_{a+1,b-1}, v_{a+1,b+1}$ are already power dominated by S_{k+1} , by rule 2, $v_{a+1,b-1}$ can be power dominated by S_{k+1} . Repeating the same process on $\{v_{a,i} | v_{a,i} \in V_{\text{even}} \text{ and } b-2 \geq i \geq 1\}$, by rule 2, $\{v_{a+1,j} | v_{a+1,j} \in V_{\text{even}} \text{ and } b-3 \geq j \geq 0\}$ can be power dominated by S_{k+1} . Since $N(v_{a-2,b}) = \{v_{a-3,b-1}, v_{a-3,b+1}, v_{a-1,b-1}, v_{a-1,b+1}\}$ and $v_{a-3,b-1}, v_{a-1,b-1}$, and $v_{a-1,b+1}$ are already power dominated by S_{k+1} , then by rule 2, $v_{a-3,b+1}$ can be power dominated by S_{k+1} . Next repeating the same process on $\{v_{i,b} | v_{i,b} \in V_{\text{even}} \text{ and } a-4 \geq i \geq 1\}$, then by rule 2, $\{v_{j,b+1} | v_{j,b+1} \in V_{\text{even}} \text{ and } a-5 \geq j \geq 0\}$ can be power dominated by S_{k+1} . Now repeating the same process on $\{v_{i,b+1} | v_{i,b+1} \in V_{\text{even}} \text{ and } a-1 \geq i \geq 1\}$ and $\{v_{i,b+2} | v_{i,b+2} \in V_{\text{even}} \text{ and } a-2 \geq i \geq 1\}$, then $\{v_{j,b+2} | v_{j,b+2} \in V_{\text{even}} \text{ and } a-2 \geq j \geq 0\}$ and $\{v_{j,b+3} | v_{j,b+3} \in V_{\text{even}} \text{ and } a-3 \geq j \geq 0\}$ is power dominated by S_{k+1} . Hence $V_{k+1} = \{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq a+1, 0 \leq j \leq b+3\}$ is power dominated by S_{k+1} . ■

Lemma 2 For $G = C_m \times C_n$, every three consecutive columns or rows must contain at least one vertex in a PDS. Which implies $\gamma_p(C_m \times C_n) \geq \max\{\lceil m/3 \rceil, \lceil n/3 \rceil\}$.

Proof. Let $L_i = \{v_{i,j} | 0 \leq j \leq n-1\}$ denote the i -th row of G for some $0 \leq i \leq m-1$. To show that $\gamma_p(G) \geq \lceil m/3 \rceil$, we show that for any PDS S of G , every three consecutive rows must have at least one vertex in S . Suppose to the contrary that $(L_i \cup L_{i-1} \cup L_{i+1}) \cap S = \emptyset$. Since $N(L_i) = L_{i-1} \cup L_{i+1}$, in order to power dominate any vertex in L_i , it must apply rule 2 on some vertices in L_{i-1} or L_{i+1} . Since every vertex in L_{i-1} and L_{i+1} has two neighbors in L_i , rule 2 can not be applied. Hence L_i can not be power dominated by S , which implies $\gamma_p(G) \geq \lceil m/3 \rceil$. $\gamma_p(G) \geq \lceil n/3 \rceil$ can be shown in a similar way. Hence we have $\gamma_p(G) \geq \max\{\lceil m/3 \rceil, \lceil n/3 \rceil\}$. ■

Since $G = C_m \times C_n$ consists of two isomorphic connected components when m and n are both even. The component containing $v_{0,0}$ is called *even component* and is denoted as G_2 and the other component is called *odd component* and is denoted as G_1 . Without loss of generality we only discuss the even component G_2 . Notice that $v_{i,j} \in V(G_2)$ if and only if $v_{i,j} \in V_{\text{even}}$. Observing that $C_m \times C_n$ for even m and odd n is isomorphic to a component of $C_m \times C_{2n}$. Lemma 3 gives the lower bound of $C_m \times C_n$ according to the value of m and n .

Lemma 3 Let $G = C_m \times C_n$. Then

- (1) for m and n are both even and $n \geq m$,
 $\gamma_p(G) \geq \max\{2\lfloor (m+n)/4 \rfloor, \lceil n/3 \rceil\}$.

- (2) for even m and odd n ,
 $\gamma_p(G) \geq \max\{\lfloor (m+2n)/4 \rfloor, \lceil 2n/3 \rceil\}$.

- (3) for m and n are both odd and $n \geq m$,
 $\gamma_p(G) \geq \max\{\lfloor (m+n+2)/4 \rfloor, \lceil n/3 \rceil\}$.

Proof. Let S be a power dominating set of G . Let M be a set of vertices of G that have been power dominated. Let $B(M)$ be the set of vertices of M that have at least a neighbor not in M . We claim that $|B(M)|$ will not increase when a new vertex adding into M is power dominated by rule 2. Suppose that w is a new vertex that is power dominated by rule 2 and w is the only neighbor not in M of some vertex v from M . Since v has all its four neighbors in M , now, v is removed from $B(M)$. It is possible that other vertices are removed from $B(M)$ if w was their fourth neighbor added to M . Except w , none of other vertex may be added into $B(M)$, so $|B(M)|$ is not increased. Since G is 4-regular, for any $S' \subseteq S$, we must have $|B(M)|$ less than or equal to $4|S'|$. According to the value of m and n , we have following three cases.

Case 1. m and n are both even.

In this case, $G = G_1 \cup G_2$. For each component of G , if we remove a vertex from S , then when $V(G) - M$ consists the vertices of a column and a row, which results the maximum $|B(M)|$ is no more than $m+n-4$. So we must have $4(|S|-1) \geq (m+n-4)$, that is $|S| \geq \lfloor (m+n)/4 \rfloor$. Hence $\gamma_p(G) \geq \max\{2\lfloor (m+n)/4 \rfloor, 2\lceil n/3 \rceil\}$.

Case 2. m is even and n is odd.

If we remove a vertex from S , then when $V(G) - M$ consists the vertices of a column and a row, which results the maximum $|B(M)|$ is no more than $m+2n-4$. So we must have $4(|S|-1) \geq (m+2n-4)$, that is $|S| \geq \lfloor (m+2n)/4 \rfloor$. Hence $\gamma_p(G) \geq \max\{\lfloor (m+2n)/4 \rfloor, \lceil 2n/3 \rceil\}$.

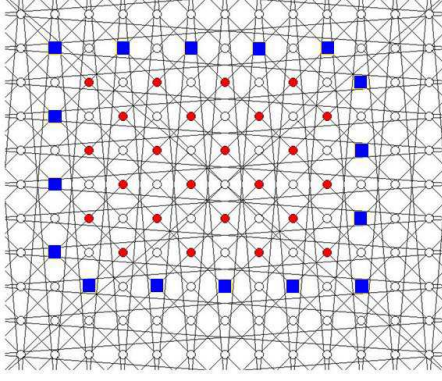
Case 3. m and n are both odd.

If we remove a vertex from S , then when $V(G) - M$ consists the vertices of V_{even} , which results the maximum $|B(M)|$ is no more than $m+n-2$. So we must have $4(|S|-1) \geq (m+n-2)$, that is $|S| \geq \lfloor (m+n+2)/4 \rfloor$. Hence $\gamma_p(G) \geq \max\{\lfloor (m+n+2)/4 \rfloor, \lceil n/3 \rceil\}$. ■

Figure 2 shows an example of $B(M)$ where the solid vertices represent the vertices in M (those have been power dominated) and square vertices represent the vertices in $B(M)$ (the boundary vertices).

Theorem 1 Let $G = C_m \times C_n$ such that m and n are both even and $n \geq m$ for $3m \leq n+6$, $\gamma_p(G) = 2\lceil n/3 \rceil$.

Proof. Let $x = \lceil n/3 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{0,0},$

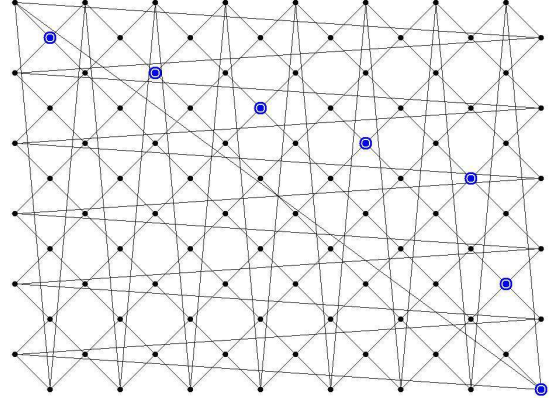

 Figure 2: A example of $B(M)$.

$v_{0,2}, v_{2,0}, v_{2,2}$ is power dominated by S . Since $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 1, $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Since $m-1 \leq x+1$ and $n-1 \leq 3x-1$, V_{even} is power dominated by S . Hence S is a PDS of G_2 , that is $\gamma_p(G_2) \leq \lceil n/3 \rceil$. Since $\gamma_p(G) = 2\gamma_p(G_2)$ by Lemma 3, $\gamma_p(G) = 2\lceil n/3 \rceil$. ■

Theorem 2 Let $G = C_m \times C_n$ such that m and n are both even and $n \geq m$ for $3m > n + 6$, $2(\lfloor (m+n)/4 \rfloor) \leq \gamma_p(G) = 2\gamma_p(G_2) \leq 2(\lceil (m+n)/4 \rceil)$.

Proof. Let $x = \lceil (3n-m)/8 \rceil$, $y = \lceil (3m-n-6)/8 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\} \cup \{v_{x+3j,3x+j-2} | 1 \leq j \leq y\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x + y$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{0,0}, v_{0,2}, v_{2,0}, v_{2,2}\}$ is power dominated by S . Since $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 1, $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Similarly, consider $\{v_{x+3t,3x+t-2} | 1 \leq t \leq y\} \subseteq S$, by Lemma 1, $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq x+3y+1, 0 \leq j \leq 3x+y-1\}$ can be power dominated by S . Since $m-1 \leq x+3y+1$ and $n-1 \leq 3x+y-1$, V_{even} can be dominated by S . Hence S is a PDS of G_2 , that is $\gamma_p(G_2) \leq \lceil (3n-m)/8 \rceil + \lceil (3m-n-6)/8 \rceil$. Since $x+y = \lfloor (m+n)/4 \rfloor$ for $m+n \neq 8k+6$ with some integer $k \geq 1$; and $x+y = \lceil (m+n)/4 \rceil$ for $m+n = 8k+6$ with some integer $k \geq 1$, by Lemma 3, $\gamma_p(G_2) = \lfloor (m+n)/4 \rfloor$ for $m+n \neq 8k+6$ and $\lfloor (m+n)/4 \rfloor \leq \gamma_p(G_2) \leq \lceil (m+n)/4 \rceil$ for $m+n = 8k+6$ with some integer $k \geq 1$. ■

Figure 3 shows an allocation of a PDS on G_2 of $C_{12} \times C_{16}$ where the vertices with a circle represent the vertices in PDS.


 Figure 3: An allocation of PDS on G_2 of $C_{12} \times C_{16}$.

Notice that $G = C_m \times C_n$ is a connected graph when either m or n is odd. Since $C_m \times C_n$ is isomorphic to a component of $C_m \times C_{2n}$ for even m and odd n , Theorem 3 and Theorem 4 come directly.

Theorem 3 Let $G = C_m \times C_n$ such that even m and odd n for $3m \leq 2n + 6$. $\gamma_p(G) = \lceil 2n/3 \rceil$.

Theorem 4 Let $G = C_m \times C_n$ such that even m and odd n for $6n \leq m + 6$. $\gamma_p(G) = \lceil m/3 \rceil$.

Theorem 5 Let $G = C_m \times C_n$ such that even m and odd n for $2n + 6 < 3m < 18n - 18$. $\lfloor (m+2n)/4 \rfloor \leq \gamma_p(G) \leq \lceil (m+2n)/4 \rceil$.

Proof. Since G is isomorphic to a component of $C_m \times C_{2n}$. By Theorem 2, $\gamma_p(G) \leq \lceil (6n-m)/8 \rceil + \lceil (3m-2n-6)/8 \rceil$. Since $\lceil (6n-m)/8 \rceil + \lceil (3m-2n-6)/8 \rceil = \lfloor (m+2n)/4 \rfloor$ for $m+2n \neq 8k+6$ with some integer $k \geq 1$; and $\lceil (6n-m)/8 \rceil + \lceil (3m-2n-6)/8 \rceil = \lceil (m+2n)/4 \rceil$ for $m+2n = 8k+6$ with some integer $k \geq 1$, by Lemma 3, $\gamma_p(G) = \lfloor (m+2n)/4 \rfloor$ for $m+2n \neq 8k+6$ and $\lfloor (m+2n)/4 \rfloor \leq \gamma_p(G) \leq \lceil (m+2n)/4 \rceil$ for $m+2n = 8k+6$ with some integer $k \geq 1$. ■

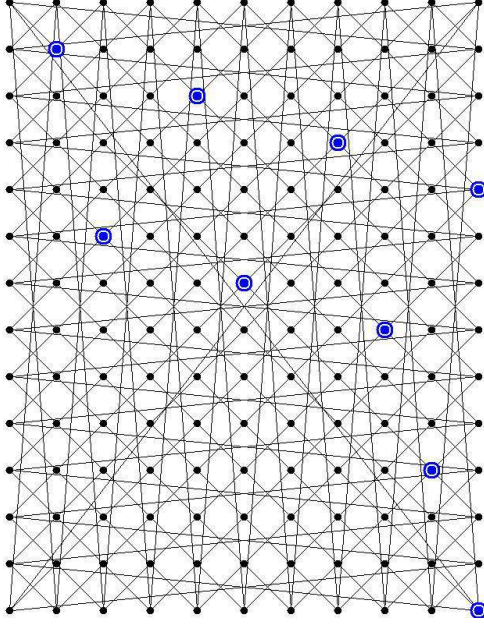
Figure 4 shows the allocation of a PDS on $C_{14} \times C_{11}$ where the vertices with a circle represent the vertices in PDS.

When m and n are both odd Lemma 4 shows that if V_{even} is power dominated by S , then $C_m \times C_n$ can be power dominated by adding just one more vertex.

Lemma 4 Let $G = C_m \times C_n$ with $n \geq m$ such that m and n are both odd. If V_{even} is power dominated by S , then $S' = S \cup \{v\}$, $v \in (\{v_{i,j} | i \in \{0,1,m-1,m-2\} \text{ or } j \in \{0,1,n-1,n-2\}\})$ can power dominate G .

Proof. Consider following three cases.

Case 1. $v \in \{v_{0,0}, v_{m-1,n-1}\}$. Since $v_{m-1,n-1}$ and $v_{0,0}$ are symmetric, without loss of generality, let

Figure 4: An allocation of PDS on $C_{14} \times C_{11}$.

$v = v_{m-1,n-1}$. By rule 1, $N(v_{m-1,n-1}) = \{v_{0,0}, v_{0,n-2}, v_{m-2,0}, v_{m-2,n-2}\}$ is power dominated by S' . By Lemma 1, $\{v_{i,0} | v_{i,0} \in V_{\text{odd}} \text{ and } 0 \leq i \leq m-4\}$ and $\{v_{0,j} | v_{0,j} \in V_{\text{odd}} \text{ and } 0 \leq j \leq n-4\}$ is power dominated by S' . Since $N(v_{1,0}) = \{v_{0,n-1}, v_{0,1}, v_{2,n-1}, v_{2,1}\}$ and $v_{0,n-1}, v_{0,1}$ and $v_{2,n-1}$ are power dominated by S' , by rule 2, $v_{2,1}$ can be power dominated by S' . Repeating the same process on $\{v_{i,0} | v_{i,0} \in V_{\text{odd}} \text{ and } 3 \leq i \leq m-2\}$, by rule 2, $\{v_{j,1} | v_{j,1} \in V_{\text{odd}} \text{ and } 4 \leq j \leq m-1\}$ can be power dominated by S' . Next repeating the same process on $\{v_{i,1} | v_{i,1} \in V_{\text{odd}} \text{ and } 0 \leq i \leq m-3\}$, by rule 2, $\{v_{j,2} | v_{j,2} \in V_{\text{odd}} \text{ and } 1 \leq j \leq m-2\}$ can be power dominated by S' . Now repeating the same process on $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq m-2, 2 \leq j \leq n-2\}$, $\{v_{a,b} | 0 \leq a \leq m-1, 0 \leq b \leq n-1\}$ can be power dominated by S' . Hence S' is a PDS of G .

Case 2. $v \in \{v_{i,j} | i \in \{0, m-1\} \text{ or } j \in \{0, n-1\}\} \cap V_{\text{odd}}$. Since $\{v_{i,j} | i \in \{0, m-1\} \text{ or } j \in \{0, n-1\}\} \cap V_{\text{odd}}$ are symmetric, without loss of generality, let $v = v_{0,n-2}$. By rule 1, $N(v_{0,n-2}) = \{v_{m-1,n-3}, v_{m-1,n-1}, v_{1,n-3}, v_{1,n-1}\}$ is power dominated by S' . By Lemma 1, $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq 1, 0 \leq j \leq n-1\}$ is power dominated by S' . Since $N(v_{2,0}) = \{v_{1,n-1}, v_{1,1}, v_{3,n-1}, v_{3,1}\}$ and $v_{1,n-1}, v_{1,1}$ and $v_{3,1}$ are power dominated by S' , by rule 2, $v_{3,n-1}$ can be power dominated by S' . Repeating the same process on $\{v_{i,0} | v_{i,0} \in V_{\text{even}} \text{ and } 4 \leq i \leq m-3\}$, by rule 2, $\{v_{j,n-1} | v_{j,n-1} \in V_{\text{odd}} \text{ and } 5 \leq j \leq m-2\}$ can be power dominated by S' . Since $N(v_{1,n-1}) = \{v_{0,n-2}, v_{0,0}, v_{2,n-2}, v_{2,0}\}$ and $v_{0,n-2}, v_{0,0}$ and $v_{2,0}$ are power dominated by S' , by rule

2, $v_{2,n-2}$ can be power dominated by S' . Repeating the same process on $\{v_{i,n-1} | v_{i,n-1} \in V_{\text{odd}} \text{ and } 3 \leq i \leq m-2\}$, by rule 2, $\{v_{j,n-2} | v_{j,n-2} \in V_{\text{odd}} \text{ and } 4 \leq j \leq m-1\}$ can be power dominated by S' . Next repeating the same process on $\{v_{i,n-2} | v_{i,n-2} \in V_{\text{odd}} \text{ and } 2 \leq i \leq m-3\}$, by rule 2, $\{v_{j,n-3} | v_{j,n-3} \in V_{\text{odd}} \text{ and } 3 \leq j \leq m-2\}$ can be power dominated by S' . Now repeating the same process on $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 1 \leq i \leq m-2, 1 \leq j \leq n-3\}$, $\{v_{a,b} | 0 \leq a \leq m-1, 0 \leq b \leq n-1\}$ can be power dominated by S' . Hence S' is a PDS of G .

Case 3. $v \in \{v_{i,j} | i \in \{1, m-2\} \text{ or } j \in \{1, n-2\}\} \cap V_{\text{odd}}$. Since $\{v_{i,j} | i \in \{1, m-2\} \text{ or } j \in \{1, n-2\}\} \cap V_{\text{odd}}$ are symmetric, without loss of generality, let $v = v_{2,n-2}$. By rule 1, $N(v_{2,n-2}) = \{v_{1,n-3}, v_{1,n-1}, v_{3,n-3}, v_{3,n-1}\}$ is power dominated by S' . By Lemma 1, $\{v_{i,j} | v_{i,j} \in V_{\text{odd}} \text{ and } 0 \leq i \leq 3, 0 \leq j \leq n-1\}$ is power dominated by S' . Now repeating the same process as Case 2, $\{v_{a,b} | 0 \leq a \leq m-1, 0 \leq b \leq n-1\}$ can be power dominated by S' . Hence S' is a PDS of G . ■

Theorem 6 Let $G = C_m \times C_n$ with $n \geq m$ such that m and n are both odd for $3m \leq n+5$. The $\gamma_p(G) = \lceil n/3 \rceil$.

Proof. Let $x = \lceil n/3 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x$. Similar to Theorem 1, $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Notice that $v_{i,j} \in V_{\text{odd}}$ when $i \geq m$. Since $2m < x+1$ and $n-1 \leq 3x-1$, G is power dominated by S . Hence S is a PDS of G , which implies $\gamma_p(G) \leq \lceil n/3 \rceil$. By Lemma 3, $\gamma_p(G) = \lceil n/3 \rceil$. ■

Theorem 7 Let $G = C_m \times C_n$ with $n \geq m$ such that m and n are both odd for $3m > n+5$. The $\gamma_p(G) = \lfloor (m+n+2)/4 \rfloor$.

Proof. Let $x = \lceil (3n-m-1)/8 \rceil$, $y = \lceil (3m-n-5)/8 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\} \cup \{v_{x+3j,3x+j-2} | 1 \leq j \leq y\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x+y$. Similar to Theorem 2, $\{v_{i,j} | v_{i,j} \in V_{\text{even}} \text{ and } 0 \leq i \leq x+3y+1, 0 \leq j \leq 3x+y-1\}$ can be power dominated by S . Since either $m-1 < x+3y+1$ and $n-1 \leq 3x+y-1$ or $m-1 \leq x+3y+1$ and $n-1 < 3x+y-1$, V_{even} is power dominated by S . Since the last vertex selected by S is $v_{x+3y,3x+y-2}$, according to the value of $x+3y$ and $3x+y-2$, by Lemma 4, G can be power dominated by S . Hence S is a PDS of G , that is $\gamma_p(G) \leq \lceil (3n-m-1)/8 \rceil + \lceil (3m-n-5)/8 \rceil$. Since $\lceil (3n-m-1)/8 \rceil + \lceil (3m-n-5)/8 \rceil = \lfloor (m+n+2)/4 \rfloor$, by Lemma 3, $\gamma_p(G) = \lfloor (m+n+2)/4 \rfloor$. ■

Figure 5 shows the allocation of a PDS of $C_{13} \times C_{15}$ where the vertices with a circle represent the vertices in PDS.

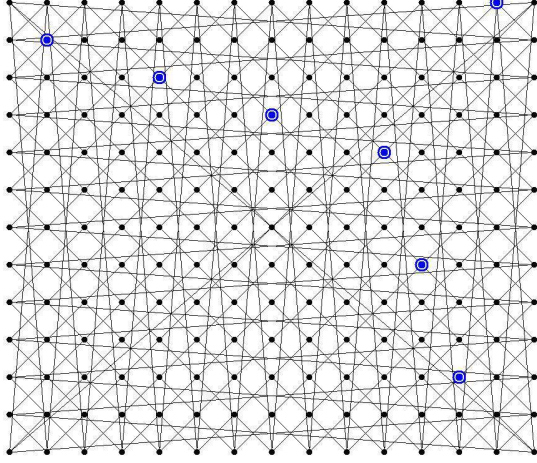


Figure 5: An allocation of PDS on $C_{13} \times C_{15}$.

4 Strong Product

The strong product $G \otimes H$ of two graphs G and H , is a graph with vertex set $V(G) \times V(H)$ and a vertex v is adjacent to a vertex u if they are equal in one coordinate and adjacent in the other or they are adjacent in both coordinates.

Similar to $C_m \times C_n$, Lemma 5 to Lemma 7 can be proved by the same way as in the proof of Lemma 1 and Lemma 3.

Lemma 5 Let $G = C_m \otimes C_n$.

- (1) If $\{v_{i,j} | 0 \leq i \leq a, 0 \leq j \leq b\}$ is power dominated by S , then $\{v_{i,j} | 0 \leq i \leq a+1, 0 \leq j \leq b+3\}$ is power dominated by $S' = S \cup \{v_{a,b+2}\}$.
- (2) If $\{v_{i,j} | 0 \leq i \leq a, 0 \leq j \leq b\}$ is power dominated by S , then $\{v_{i,j} | 0 \leq i \leq a+3, 0 \leq j \leq b+1\}$ is power dominated by $S' = S \cup \{v_{a+2,b}\}$.

Lemma 6 For $G = C_m \otimes C_n$, every three consecutive columns or rows must have at least one vertex in a PDS. Which implies $\gamma_p(G) \geq \max\{\lceil m/3 \rceil, \lceil n/3 \rceil\}$.

By using a similar argument of the boundary set, the lower bound of the power domination number of $C_m \otimes C_n$ is shown in Lemma 7.

Lemma 7 Let $G = C_m \otimes C_n$ for $n \geq m$. Then $\gamma_p(G) \geq \max\{\lceil n/3 \rceil, \lfloor (m+n)/4 \rfloor\}$.

Theorem 8 shows the power domination number of $G = C_m \otimes C_n$ such that $3m \leq n+7$ for $n \geq m$.

Theorem 8 Let $G = C_m \otimes C_n$ such that $3m \leq n+7$ for $n \geq m$. Then $\gamma_p(G) = \lceil n/3 \rceil$.

Proof. Let $x = \lceil n/3 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 5, $\{v_{i,j} | 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Since $m-1 \leq x+1$ and $n-1 \leq 3x-1$, G is power dominated by S . Hence S is a PDS of G with the cardinality $\lceil n/3 \rceil$. By Lemma 7 $\gamma_p(G) = \lceil n/3 \rceil$. ■

Theorem 9 gives the sharp upper bound of power domination number for connected case of even m and odd n . The upper bound will no more than $\lfloor (m+n)/4 \rfloor + 1$ which is one more than the lower bound.

Theorem 9 Let $G = C_m \otimes C_n$ such that $3m > n+7$ for $n \geq m$. Then $\lfloor (m+n)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n)/4 \rfloor + 1$.

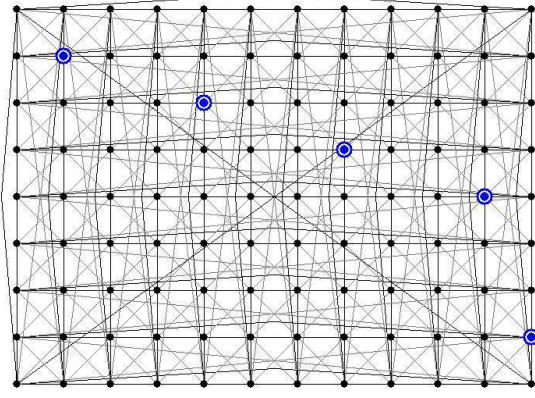
Proof. Let $x = \lceil (3n-m)/8 \rceil$, $y = \lceil (3m-n-7)/8 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\} \cup \{v_{x+3j,3x+j-2} | 1 \leq j \leq y\}$ where the first subscript sum is taken modulo m and the second one is taken modulo n . Note that $|S| = x+y$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 5, $\{v_{i,j} | 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Similarly, consider $\{v_{x+3t,3x+t-2} | 1 \leq t \leq y\} \subseteq S$, by Lemma 5, $\{v_{i,j} | 0 \leq i \leq x+3y+1, 0 \leq j \leq 3x+y-1\}$ can be power dominated by S . Since $m-1 \leq x+3y+1$ and $n-1 \leq 3x+y-1$, G can be dominated by S . That is S is a PDS of G with the cardinality $x+y$. Let $3n-m = 8q_1 + r_1$ and $3m-n-7 = 8q_2 + r_2$. If $r_1 \geq 1$, $r_2 \geq 1$ and $r_1 + r_2 < 9$, $\gamma_p(G) \leq \lfloor (m+n)/4 \rfloor + 1$, else $\gamma_p(G) = \lfloor (m+n)/4 \rfloor$. ■

Figure 6 shows the allocation of a PDS of $C_9 \otimes C_{12}$ where the vertices with a circle represent the vertices in PDS.

Similar to $C_m \times C_n$, Lemma 8 to Lemma 10 can be proved by the same way as in the proof of Lemma 1 and Lemma 3.

Lemma 8 Let $G = P_m \otimes C_n$.

- (1) If $\{v_{i,j} | 0 \leq i \leq a, 0 \leq j \leq b\}$ is power dominated by S , then $\{v_{i,j} | 0 \leq i \leq a+1, 0 \leq j \leq b+3\}$ is power dominated by $S' = S \cup \{v_{a,b+2}\}$.
- (2) If $\{v_{i,j} | 0 \leq i \leq a, 0 \leq j \leq b\}$ is power dominated by S , then $\{v_{i,j} | 0 \leq i \leq a+3, 0 \leq j \leq b+1\}$ is power dominated by $S' = S \cup \{v_{a+2,b}\}$.


 Figure 6: An allocation of PDS on $C_9 \otimes C_{12}$.

Lemma 9 For $G = P_m \otimes C_n$, every three consecutive columns or rows must have at least one vertex in a PDS. Which implies $\gamma_p(G) \geq \max\{\lceil m/3 \rceil, \lceil n/3 \rceil\}$.

By using a similar argument of the boundary set, the lower bound of the power domination number of $P_m \otimes C_n$ is shown in Lemma 10.

Lemma 10 Let $G = P_m \otimes C_n$. Then $\gamma_p(G) \geq \max\{\lceil n/3 \rceil, \lfloor (m+n+1)/4 \rfloor\}$ for $n \geq m$.

Theorem 10 shows the power domination number of $G = P_m \otimes C_n$ such that $3m \leq n+7$ and $n \geq 3$.

Theorem 10 Let $G = P_m \otimes C_n$ such that $3m \leq n+7$ and $n \geq 3$. Then $\gamma_p(G) = \lceil n/3 \rceil$.

Proof. Let $x = \lceil n/3 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\}$ where the second subscript sum is taken modulo n and if $x > m-1$ then take $v_{m-1,3x-2}$ instead of $v_{x,3x-2}$. Note that $|S| = x$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Since $m-1 \leq x+1$ and $n-1 \leq 3x-1$, G is power dominated by S . Hence S is a PDS of G with the cardinality $\lceil n/3 \rceil$. By Lemma 10 $\gamma_p(G) = \lceil n/3 \rceil$. ■

Theorem 11 Let $G = P_m \otimes C_n$ such that $3n \geq 3m > n+7$. Then $\lfloor (m+n+1)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$.

Proof. Let $x = \lceil (3n-m)/8 \rceil$, $y = \lceil (3m-n-7)/8 \rceil$ and $S = \{v_{i+1,3i+1} | 0 \leq i < x\} \cup \{v_{x+3j,3x+j-2} | 1 \leq j \leq y\}$ where the second subscript sum is taken modulo n and if $x+3y > m-1$ then take $v_{m-1,3x+y-2}$ instead of $v_{x+3y,3x+y-2}$. Note that $|S| = x+y$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is

power dominated by S , consider $\{v_{t+1,3t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq x+1, 0 \leq j \leq 3x-1\}$ can be power dominated by S . Similarly, consider $\{v_{x+3t,3x+t-2} | 1 \leq t \leq y\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq x+3y+1, 0 \leq j \leq 3x+y-1\}$ can be power dominated by S . Since $m-1 \leq x+3y+1$ and $n-1 \leq 3x+y-1$, G can be dominated by S . That is S is a PDS of G with the cardinality $x+y$. Let $3n-m = 8q_1+r_1$ and $3m-n-7 = 8q_2+r_2$. If $r_1 \geq 1$, $r_2 \geq 1$ and $r_1+r_2 = 5$, $\gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$, else $\gamma_p(G) = \lfloor (m+n+1)/4 \rfloor$. ■

Theorem 12 shows the power domination number of $G = P_m \otimes C_n$ such that $3n \leq m+7$ and $m \geq 2$.

Theorem 12 Let $G = P_m \otimes C_n$ such that $3n \leq m+7$ and $m \geq 2$. Then $\gamma_p(G) = \lceil m/3 \rceil$.

Proof. Let $x = \lceil m/3 \rceil$ and $S = \{v_{3i+1,i+1} | 0 \leq i < x\}$ where the second subscript sum is taken modulo n and if $3x-2 > m-1$ then take $v_{m-1,x}$ instead of $v_{3x-2,x}$. Note that $|S| = x$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{3t+1,t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq 3x-1, 0 \leq j \leq x+1\}$ can be power dominated by S . Since $m-1 \leq 3x-1$ and $n-1 \leq x+1$, G is power dominated by S . Hence S is a PDS of G with the cardinality $\lceil m/3 \rceil$. By Lemma 10 $\gamma_p(G) = \lceil m/3 \rceil$. ■

Theorem 13 Let $G = P_m \otimes C_n$ such that $3m \geq 3n > m+7$. Then $\lfloor (m+n+1)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$.

Proof. Let $x = \lceil (3m-n)/8 \rceil$, $y = \lceil (3n-m-7)/8 \rceil$ and $S = \{v_{3i+1,i+1} | 0 \leq i < x\} \cup \{v_{3x+j-2,x+3j} | 1 \leq j \leq y\}$ where the second subscript sum is taken modulo n and if $3x+y-2 > m-1$ then take $v_{m-1,x+3y}$ instead of $v_{3x+y-2,x+3y}$. Note that $|S| = x+y$. Since $v_{1,1} \in S$, by rule 1, $N(v_{1,1}) = \{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S . Since $\{v_{i,j} | 0 \leq i, j \leq 2\}$ is power dominated by S , consider $\{v_{3t+1,t+1} | 1 \leq t < x\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq 3x-1, 0 \leq j \leq x+1\}$ can be power dominated by S . Similarly, consider $\{v_{3x+t-2,x+3t} | 1 \leq t \leq y\} \subseteq S$, by Lemma 8, $\{v_{i,j} | 0 \leq i \leq 3x+y-1, 0 \leq j \leq x+3y+1\}$ can be power dominated by S . Since $m-1 \leq 3x+y-1$ and $n-1 \leq x+3y+1$, G can be dominated by S . That is S is a PDS of G with the cardinality $x+y$. Let $3n-m-7 = 8q_1+r_1$ and $3m-n = 8q_2+r_2$. If $r_1 \geq 1$, $r_2 \geq 1$ and $r_1+r_2 = 5$, $\gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$, else $\gamma_p(G) = \lfloor (m+n+1)/4 \rfloor$. ■

Figure 7 shows the allocation of a PDS of $P_{12} \otimes C_9$ where the vertices with a circle represent the vertices in PDS.

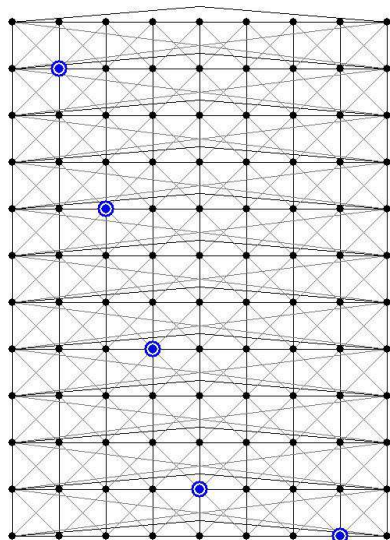


Figure 7: An allocation of PDS on $P_{12} \otimes C_9$.

5 Conclusion

Power domination is very interesting problem which grab a lot of attention in practically and theoretically as mentioned in section two.

For the tensor product graphs, Dorbec et al. [5] determined the power domination number for all tensor product of paths except for the odd component of the tensor product of two odd paths (which they provided an upper bound). They also computed the same for the strong product of paths. Varghese et al. [19] provided a formula solution for the tensor product of an even path with a cycle. In this paper we establish the results which complete the tensor product and strong product of cycle and path. The results of tensor product and strong product graphs are summarized in Table 1 and Table 2 respectively.

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Table 1: The power domination number of tensor product graphs.

Graph	$\gamma_p(G)$
$G = P_m \times P_n$ (by Dorbec et al. 2008 [5])	(1) For even n and either odd m or $m \geq n$, $\gamma_p(G) = 2\lceil n/4 \rceil$. (2) For m and n are both odd, $\gamma_p(G) = \max\{\lceil n/4 \rceil, \lceil (m+n)/6 \rceil\} + \max\{\lceil (n-2)/4 \rceil, \lceil (m+n-2)/6 \rceil\}$.
$G = C_m \times P_n$ (by Varghese et al. 2011 [20])	(1) For even m , $\gamma_p(G) = 2\lceil n/3 \rceil$. (2) For odd m , $\gamma_p(G) = \lceil n/3 \rceil$.
$G = C_m \times C_n$ (this paper)	For m and n are both even, (1) when $3m \leq n + 6$, $\gamma_p(G) = 2\gamma_p(G_2) = 2(\lceil n/3 \rceil)$. (2) when $3m > n + 6$ and $m + n = 8k + 6$ for some integer $k \geq 1$, $2(\lfloor (m+n)/4 \rfloor) \leq \gamma_p(G) \leq 2(\lceil (m+n)/4 \rceil)$, otherwise $\gamma_p(G) = 2(\lceil (m+n-2)/4 \rceil)$.
	For even m and odd n , (1) when $3m \leq 2n + 6$, $\gamma_p(G) = \lceil 2n/3 \rceil$. (2) when $6n \leq m + 6$, $\gamma_p(G) = \lceil m/3 \rceil$. (3) when $2n + 6 < 3m < 18n - 18$ and $m + 2n = 8k + 6$ for some integer $k \geq 1$, $\lfloor (m+2n)/4 \rfloor \leq \gamma_p(G) \leq \lceil (m+2n)/4 \rceil$, otherwise $\gamma_p(G) = \lceil (m+2n-2)/4 \rceil$.
	For m and n are both odd, (1) when $3m \leq n + 5$, $\gamma_p(G) = \lceil n/3 \rceil$. (2) when $3m > n + 5$, $\gamma_p(G) = \lceil (m+n)/4 \rceil$.

Table 2: The power domination number of strong product graphs.

Graph	$\gamma_p(G)$
$G = (P_m \otimes P_n)$ (by Dorbec et al. 2008 [5])	$\gamma_p(G) = \max\{\lceil n/3 \rceil, \lceil (m+n-2)/4 \rceil\}$
$G = (P_m \otimes C_n)$ (this paper)	(1) When $3m \leq n + 7$ and $n \geq 3$, $\gamma_p(G) = \lceil n/3 \rceil$. (2) When $3n \geq 3m > n + 7$, $\gamma_p(G) = \lfloor (m+n+1)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$. (3) When $3n \leq m + 7$ and $m \geq 2$, $\gamma_p(G) = \lceil m/3 \rceil$. (4) When $3m \geq 3n > m + 7$, $\gamma_p(G) = \lfloor (m+n+1)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n+1)/4 \rfloor + 1$.
$G = (C_m \otimes C_n)$ (this paper)	(1) When $3m \leq n + 7$ and $n \geq m$, $\gamma_p(G) = \lceil n/3 \rceil$. (2) When $3n \geq 3m > n + 7$, $\gamma_p(G) = \lfloor (m+n)/4 \rfloor \leq \gamma_p(G) \leq \lfloor (m+n)/4 \rfloor + 1$.

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